

Cleomene Nelli

Computer Engineer

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Education

Trinity College Dublin, Dublin, Ireland

September 2021 – May 2026

Integrated Masters in Computer Engineering (MEng)

- **Relevant Coursework:** Deep Learning (Python), Data Structures & Algorithms, Information Management (MySQL), Optimization Algorithms for Data Analysis, Security and Privacy

Politecnico di Torino, Torino, Italy

February 2025 – July 2026

Erasmus+ Exchange in Electronic Engineering Masters Program

- **Relevant Coursework:** Distributed Architectures for Big Data (Hadoop, Spark), Optimization of Digital Systems

Technical Skills

- **Programming:** Python, C++, C, Java, SQL, HTML/CSS, Verilog/VHDL, MATLAB, Assembly
- **OS:** Mac, Windows, Ubuntu/Linux
- **Development:** Git, Kubernetes, Docker, Command Line, Jira
- **Hardware:** Custom PC Builds, Laptop repair
- **Soft Skills:** Creative problem solving, time management, technical communication, teamwork

Work Experience

Tala, Los Angeles, CA

June 2024 – August 2024

Software Engineer Intern

- Built and updated APIs in Java and Scala, improving data access and deprecating legacy endpoints.
- Used Docker and MySQL to test and validate services.
- Deployed code with GitHub and Jenkins, following Agile practices and peer code reviews.

Projects

More at cleomenenelli.us & github.com/CleNel

Extending Kubernetes Nodes to the Edge | Kubernetes, WireGuard, Containerd, Azure

- Integrated a developer laptop as a live Kubernetes node into an AKS cluster using Feather, a lightweight Virtual Kubelet-based container runtime
- Engineered transparent bi-directional networking between the edge device and cluster by configuring a Tailscale subnet router, DNS forwarding via CoreDNS, and system-level IP routing
- Evaluated prototype performance through connectivity, latency, and load testing, demonstrating seamless edge-to-cluster service communication without pod-level modification

Super-Resolution Pipeline | Python, Tensorflow (Keras)

- Built a deep learning pipeline to upscale diffusion model outputs from 32x32 to 128x128
- Implemented a custom perceptual loss combining pixel-level and VGG19 feature loss
- Designed and trained a convolutional neural network with up-sampling layers
- Learned how to use SSIM for validation and performance tracking